15. SETTING UP THE 3RD CHAMBER

DURATION : 06:17

STUDY SHEET

COURSE SUMMARY

This video addresses the importance of the third embryonic chamber in the context of differential cell growth. It explores the role of the blastocoel, the amniotic cavity, the endocyst, and the growth algorithm that conditions the formation of different cavities: the vitelline cavity, the exocoelomic cavity, and the amniotic cavity. Emphasis is placed on fluid dynamics and the impact of cell divisions, thus merging embryological understanding with potential applications in osteopathy and biodynamics.

SETTING UP THE 3RD CHAMBER

We are approaching a crucial moment: the appearance of the **third chamber**.

The fundamental principle to remember here is **differential growth rate**. Structures do not all grow at the same rate.

DYNAMICS OF THE BLASTOCOEL AND THE APPEARANCE OF CAVITIES

We are in the **blastocoel**, within a totality. Observing the small cavity following implantation in the mucosa, we note that the cells facing this cavity will form, according to Blechschmidt, an **endocyst**.

However, within this endocyst, some cells turn towards the future **amniotic cavity**. These cells will form the endocyst or the **epiblast**.

On the other other side, another large cavity is visible. This is the endocyst, or for us, the **endoblast**. So we have **epiblast** and **endoblast**.

THE DIFFERENTIAL GROWTH ALGORITHM

The algorithm determining this differentiation is the proximity of the trophic supply. Which cells, among all those present, are closest to the nutrients and will grow the fastest?

In black, we observe a **differentiated growth rate** within the embryo itself, between the central cells and those of the periphery. Growth between some areas remains similar, but between others, there is tension, a stretching.

This indicates that the blastocoel is already linked to the amniotic cavity, to the fluid, to the digestive system. We observe a tearing, a progressive separation. This stretching will amplify.

Normally, without growth, the image would return to the centre. However, the differential growth rate between the centre and the periphery of the embryo generates this dynamic.

RECONFIGURATION OF CAVITIES AND NAMES

This system creates a layer that maintains its density thanks to a **membrane**, called the **blastocoel membrane**.

At this stage, the entire blastocoel changes its name:

- The centre becomes the **vitelline cavity**.
- The periphery becomes the **external coelom**.

The centre does not seem to grow proportionally, while all around, growth is intense, which causes tearing. This is how the coelom, previously mentioned, becomes the vitelline cavity.

Around it, another cavity appears. Initially, it takes the form of an **arachnoid network**. Then, through growth, this network eliminates itself to form a cavity: the **external coelom**.

This arachnoid network, initially, holds the structures. But at some point, it loosens, leaving an **exudate** which becomes the external coelom. This cavity will become the **peritoneal fluid cavity**, intra-peritoneal.

SYNTHESIS OF FLUID CAVITIES

We end up with three fluid cavities:

- The **amniotic cavity** (represented here in yellow).
- The **vitelline cavity**.
- The **external coelom**.

Between these different cavities is the **embryonic disc**.

- The cells of the endocyst oriented towards the amniotic cavity will form, in our language, the **epiblast**.
- The cells oriented towards the vitelline cavity will form the **endoblast**.

THE ROLE OF CELL DIVISION AND FLUIDS

Each **cell division** leads to water loss and an increase in membrane surface. This can manifest as small exudates or a complete exudate.

The term "exudate" also refers to growth. Imagine the number of cells that appear, each originating from a previous cell.

Each cell division causes a slight loss. The growth rate is such that a considerable volume of fluid appears. Initially, the cells are held by a **fibrous network**. But over time, this fibrous network degrades, giving way to the cavities.

This is what is called the **primitive arachnoid network** of the external coelom. It is not cells breaking, but rather a massive fluid absorption, stronger from the outside in, in the periphery. One can imagine it as a fluid-rich environment where fluids are absorbed in large quantities.